

Tutorial 1

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Problem 1 Binder Selection



- The latitude at a location where a high-speed rural road is to be located is 41° . The seven-day average high temperature is 35°C and the low air temperature is -10°C . The standard deviation for both the high and low temperature is $\pm 1^\circ\text{C}$. Determine a suitable binder that could be used if the depth of the asphalt pavement is 155 mm and the expected traffic is 12×10^6 ESA.

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$$T_{20mm} = 0.9545 * (T_{air} - 0.00618 * Lat^2 + 0.2289 * Lat + 42.2) - 17.78$$

$$T_{20mm} = 0.9545 * (35 - 0.00618 * 41^2 + 0.2289 * 41 + 42.2) - 17.78 = 54.95 \approx 55^{\circ}C$$

$$T_{min} = 0.859 * T_{air} + 1.7$$

$$T_{min} = 0.859 * -10 + 1.7 = -6.9 \approx -7^{\circ}C$$

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- For 50% reliability and Normal section of the road
- PG58-16
- For slow traffic section
- PG64-10
- For standing traffic
- PG70-10

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- For 98% reliability
- Max 7-day temp 57
- Min Tem=-9
- PG58-16
- For slow traffic section
- PG64-10
- For standing traffic
- PG70-10

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Superpave Binder Grades

High Temperature Grade	Low Temperature Grade
PG46-	44,40,46
PG52-	10,16,22,28,34,40,46
PG58-	16,22,28,34,40
PG64-	10,16,22,28,34,40
PG70-	10,16,22,28,34,40
PG76-	10,16,22,28,34
PG82-	10,16,22,28,34

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Problem 2

In a quality control exercise, the engineer in charge cut an asphalt core 100 mm diameter and 65 mm deep. The total mass of the core is 1200 g. The engineer ran an extraction test in which bitumen was separated from aggregates, the mass of the aggregate was found to be 1140 g. The specific gravity of the bitumen is 1.05 and the specific gravity of the aggregate is 2.75. If the bitumen absorption is ignored; determine:

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1. The bitumen content.
2. What is percent of the aggregates in the total mix (%Ps)?
3. What is effective specific gravity of aggregates
4. What is bulk specific gravity of the compacted mix?
5. What is the maximum theoretical specific gravity?
6. What is the percent of the voids in the mineral aggregate (%VMA)?

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7. What is the percent of the voids in the total mix (%VTM)?
8. What is the percent of the voids filled with bitumen (%VFB)
9. For a dense graded mix, comment on the calculated air voids.
10. For this dense graded mix, do you expect any problems? What are these problems?

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Solution

$$V_{mb} = \frac{1}{4} * \pi * (10)^2 * 6.5 = 510.51 \text{ cm}^2$$

$$m_s = 1140 \text{ g}$$

$$m_b = 60 \text{ g}$$

$$p_s = 1140 / 1200 * 100 = 95\%$$

$$P_b = 60 / 1200 * 100 = 5.0\%$$

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$$G_{mb} = \frac{1200}{510.51} = 2.3506$$

$$G_{sb} = G_{se} \quad \text{Because } P_{ba} = 0$$

$$G_{mm} = \frac{100}{\frac{95}{2.75} + \frac{5.0}{1.05}} = 2.54405$$

$$\%VTM = \left(1 - \frac{2.3506}{2.54405}\right) * 100 = 7.604\%$$

$$\%VMA = \left(100 - \frac{95 * 2.3506}{2.75}\right) = 18.797\%$$

$$VFB = \frac{18.797 - 7.604}{18.797} * 100 = 59.55\%$$

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Problem 3

- In a quality control exercise, the engineer in charge cut an asphalt core 100 mm diameter and 65 mm deep. The total mass of the core is 1200 g. The engineer ran an extraction test in which bitumen was separated from aggregates, the mass of the aggregate was found to be 1140 g. The specific gravity of the bitumen is 1.05 and the specific gravity of the aggregate is 2.75. If the bitumen absorption is 0.9%,

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1. The bitumen content.
2. What is percent of the aggregates in the total mix (%Ps)?
3. What is effective specific gravity of aggregates
4. What is bulk specific gravity of the compacted mix?
5. What is the maximum theoretical specific gravity?
6. What is the percent of the voids in the mineral aggregate (%VMA)?

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7. What is the percent of the voids in the total mix (%VTM)?
8. What is the percent of the voids filled with bitumen (%VFB)
9. For a dense graded mix, comment on the calculated air voids.
10. For this dense graded mix, do you expect any problems? What are these problems?

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$$m_s = 1140g$$

$$m_b = 60g$$

$$p_s = 1140 / 1200 * 100 = 95\%$$

$$P_b = 60 / 1200 * 100 = 5.0\%$$

$$P_{ba} = \frac{G_{se} - G_{sb}}{G_{se} * G_{sb}} * G_b * 100$$

$$0.9 = \frac{G_{se} - 2.75}{2.75 * G_{se}} * 1.05 * 100$$

$$G_{se} = 2.81639$$

Note G_{se} is greater than G_{sb}

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$$G_{mb} = \frac{1200}{510.51} = 2.3506$$

$$G_{mm} = \frac{100}{\frac{95}{2.81639} + \frac{5.0}{1.05}} = 2.59789$$

$$\%VTM = \left(1 - \frac{2.3506}{2.59787}\right) * 100 = 9.52\%$$

$$\%VMA = \left(100 - \frac{95 * 2.3506}{2.75}\right) = 18.797\%$$

$$VFB = \frac{18.797 - 9.52}{18.797} * 100 = 49.35\%$$

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DESIGN ESA (million)	Adjustment to Binder PG Grade		
	Traffic Load Rate		
	Standing	Slow	Standard
< 0.3	-	-	-
0.3 to < 3	2	1	-
3 to < 10	2	1	-
10 to < 30	2	1	-
<u>≥ 30</u>	2	1	1

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Problem 4

- After two years of traffic, cores were recovered from the roadway which has severe rutting and bleeding. The bulk specific gravity and the maximum theoretical specific gravity were measured in these cores and are as follows:
- Bulk specific gravity = 2.487
- Maximum theoretical specific gravity = 2.561
- Calculate the air voids
- If the design air void content was 4%, explain what effect the calculated air voids had on the rutting and bleeding noted.

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$$\%VTM = \left(1 - \frac{G_{mb}}{G_{mm}}\right) * 100$$

$$\%VTM = \left(1 - \frac{2.487}{2.561}\right) * 100 = 2.9\%$$

Rutting and bleeding will occur

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Problem 5

- As part of mix design, a laboratory-compacted cylindrical asphalt specimen is weighed for determination of bulk specific gravity. The following numbers are obtained:
 - Dry Mass in air = 1264.7g
 - Mass when submerged in water = 723.9g
 - Mass of Saturated Surface Dry (SSD) = 1271.9g
- What is the bulk specific gravity of the compacted specimen (G_{mb})?
- If the maximum theoretical specific gravity of the specimen (G_{mm}) is 2.531, what would be the air void content of the specimen in percent?

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Solution

$$G_{mb} = \frac{A}{B - C}$$

$$G_{mb} = \frac{1264.7}{1271.9 - 7239} = 2.30785$$

$$\%VTM = \left(1 - \frac{2.30785}{2.531}\right) * 100 = 8.82\%$$

Ravelling and oxidation will occur.